MGU-UGP(Honours)Regular Examination, October 2025

MG3DSCPHY201- ESSENTIAL MATHEMATICS FOR PHYSICS

Time:-2 Hour **Marks:- 70**

Part A

Answer any Ten, Each carries 3 marks **(10×3=30)**

1. **Question:-** State Bessel's inequality and explain.
Answer:- Let $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_M, \ldots, \mathbf{e}_N\}$ form an orthonormal basis in N- dimensional vector space(inner product space) H. Then for any vector $\mathbf{x} \in H$,

$$\sum_{i=1}^{M} |\langle \mathbf{x}|\mathbf{e}_i\rangle|^2 \leq \|x\|^2.$$

Where the equality holds if $M = N$. If $M < N$ Then inequality results.

2 mark

$$\mathbf{x} = \sum_{i=1}^{N} \langle \mathbf{x}|\mathbf{e}_i\rangle \mathbf{e}_i \quad \text{and} \quad \|x\|^2 = \sum_{i=1}^{N} |\langle \mathbf{x}|\mathbf{e}_i\rangle|^2$$

$$\text{Let} \quad x_M = \sum_{i=1}^{M} \langle \mathbf{x}|\mathbf{e}_i\rangle \mathbf{e}_i \quad \text{and} \quad \|x_M\|^2 = \sum_{i=1}^{M} |\langle \mathbf{x}|\mathbf{e}_i\rangle|^2$$

x_M be the projection of the vector $\mathbf{x}$ onto a subspace spanned by the basis set $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_M\}$, which is a subset of $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_M, \ldots \mathbf{e}_N\}$. Then, $\|x\|^2 = \|x_M\|^2 + \|x - x_M\|^2$(Pythagorean theorem).

Or any other Correct/Sensible Answer/Explanation - 1 mark

2. **Question:- The equation of motion of a particle is given by $\mathbf{r}(t) = 2t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ Find its velocity and acceleration vectors at $t = 2s$**
Answer:- The Velocity(Velocity vector) is the time derivative of position: $\mathbf{v}(t) = \dfrac{d\mathbf{r}}{dt}$

$$\begin{aligned} \frac{d[\mathbf{r}(t)]}{dt} &= \frac{d}{dt}(2t)\mathbf{i} + \frac{d}{dt}\left(t^2\right)\mathbf{j} + \frac{d}{dt}\left(t^3\right)\mathbf{k} \\ \therefore \mathbf{v}(t) &= 2\mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k} \end{aligned}$$

1 mark

Acceleration(Acceleration vector) is the time derivative of velocity: $\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}$

$$\begin{aligned} \frac{d}{dt}(\mathbf{v}(t)) &= \frac{d}{dt}(2)\mathbf{i} + \frac{d}{dt}(2t)\mathbf{j} + \frac{d}{dt}\left(2t^2\right)\mathbf{k} \\ \therefore \mathbf{a}(t) &= 0\mathbf{i} + 2\mathbf{j} + 6t\mathbf{k} \end{aligned}$$

1 mark

At t = 2s

$$\begin{aligned} \mathbf{v}(2) &= 2\mathbf{i} + 2(2)\mathbf{j} + 3(2)^2\mathbf{k} = 2\mathbf{i} + 4\mathbf{j} + 12\mathbf{k} \\ \mathbf{a}(2) &= 0\mathbf{i} + 2\mathbf{j} + 6(2)\mathbf{k} = 2\mathbf{j} + 12\mathbf{k} \end{aligned}$$

1 mark for $\mathbf{v}(2)$ or $\mathbf{a}(2)$.

3. **Question:- Define orthonormality of base vectors. Why is it important in vector spaces?**
   Answer:- Let $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ be a set of basis vectors in a vector space(an inner product space)H. These vectors form an orthonormal basis

$$\langle \mathbf{e}_i | \mathbf{e}_j \rangle = \delta_{ij}, \quad \text{where} \quad \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

OR

- **Orthogonality:** $\langle \mathbf{e}_i | \mathbf{e}_j \rangle = 0$ for all $i \neq j$, meaning that the projection of one vector onto the other is zero, and therefore the vectors are linearly independent.
- **Normalization:** $\langle \mathbf{e}_i | \mathbf{e}_i \rangle = 1$ for all i, meaning that each vector has a unit length.

Or any other Correct/Sensible Answer - 2 mark

i. The orthonormality condition ensures that any vector $\mathbf{a} \in H$ can be expressed uniquely as

$$\mathbf{a} = \sum_{i=1}^{n} \langle \mathbf{a} | \mathbf{e}_i \rangle \mathbf{e}_i$$

where the coefficients $\langle \mathbf{a}, \mathbf{e}_i \rangle$ represent the components of $\mathbf{a}$ along the respective directions of the basis vectors.

ii. Using the orthonormality condition inner products and norms can be simplified to

$$\langle \mathbf{a} | \mathbf{b} \rangle = \sum_{i=1}^{n} a_i b_i \quad \text{and} \quad \|\mathbf{a}\|^2 = \sum_{i=1}^{n} |a_i|^2$$

where $a_i = \langle \mathbf{a} | \mathbf{e}_i \rangle$ and $b_i = \langle \mathbf{b} | \mathbf{e}_i \rangle$. Hence, orthonormal bases greatly simplify vector decomposition, projections, and its computations.

Any one or any other Correct/Sensible Answer -1 mark

4. **Question:-** **Define an upper triangular matrix and give any one property of it.**
Answer:- An **upper triangular matrix** is a square matrix in which all the elements below the principal diagonal(main diagonal) are zero.

OR

Mathematically, for an $n \times n$ matrix $A = [a_{ij}]$, $a_{ij} = 0 \quad$ for all $i > j$

Or any other Correct/Sensible Answer - 2 mark

OR

Example:- Any accurate example of an upper triangular matrix - $1\frac{1}{2}$ mark

Properties:

I. The determinant of an upper triangular matrix is equal to the product of its principal diagonal (main diagonal) elements. $\det(A) = a_{11} \times a_{22} \times \cdots \times a_{nn}$

II. For an upper triangular matrix, the eigenvalues are exactly the entries on its principal diagonal(main diagonal).

III. The inverse of a non-singular upper triangular matrix is also an upper triangular matrix.

Any one Or any other Correct/Sensible Answer - 1 mark

5. **Question:-** $A = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ Calculate $D = 2A + B - C$

Answer:-

$$\begin{aligned} 2A &= 2 \times \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 0 & 2 \end{bmatrix} \\ 2A + B &= \begin{bmatrix} -2 & 4 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} (-2+3) & (4+(-1)) \\ (0+1) & (2+2) \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \\ D &= (2A + B) - C = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} (1-1) & (3-1) \\ (1-1) & (4-1) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 2 \\ 0 & 3 \end{bmatrix} \end{aligned}$$

3 mark

6. **Question:-** $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ Verify whether the matrix **A** is orthogonal.
Answer:- A matrix A is orthogonal if $A^T A = I$

OR

A matrix A is orthogonal if $A^T = A^{-1}$

1 mark

$$A^T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

1 mark

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \det(A) = (1)(-1) - (0)(0) = -1$$

$$\therefore A^{-1} = A^T \Rightarrow A \text{ is orthogonal.}$$

OR

$$A^T A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Since $A^T A = I$, the matrix A is an Orthogonal Matrix.

1 mark

7. **Question:-** Write down the expression for the Curl of a vector function in Cylindrical coordinates (r, θ, z).

Answer:- Let $\vec{A} = A_r, \hat{\mathbf{e}}_r + A_\theta \hat{\mathbf{e}}_\theta + A_z \hat{\mathbf{e}}_z$ Then the curl of $\vec{A}$ in cylindrical coordinates is given by:

$$\nabla \times \vec{A} = \frac{1}{r} \begin{vmatrix} \hat{\mathbf{e}}_r & r\hat{\mathbf{e}}_\theta & \hat{\mathbf{e}}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ A_r & rA_\theta & A_z \end{vmatrix}$$

OR

$$\begin{aligned} \nabla \times \vec{A} &= \frac{1}{r}\left[\left(\frac{\partial A_z}{\partial \theta} - \frac{\partial (rA_\theta)}{\partial z} \right) \hat{\mathbf{e}}_r + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \hat{\mathbf{e}}_\theta + \left(\frac{\partial (rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right) \hat{\mathbf{e}}_z \right] \\ &= \left(\frac{1}{r}\frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z} \right) \hat{\mathbf{e}}_r + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \hat{\mathbf{e}}_\theta + \frac{1}{r}\left(\frac{\partial (rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right) \hat{\mathbf{e}}_z. \end{aligned}$$

Correct/Sensible Answer - 3 mark

8. **Question:-** Write down the mathematical expression for the Gradient of a scalar $\nabla\phi(q_1, q_2, q_3)$ field in general curvilinear coordinate system.

Answer:-

$$\nabla\phi(q_1, q_2, q_3) = \frac{1}{h_1}\frac{\partial \phi}{\partial q_1}\hat{\mathbf{e}}_1 + \frac{1}{h_2}\frac{\partial \phi}{\partial q_2}\hat{\mathbf{e}}_2 + \frac{1}{h_3}\frac{\partial \phi}{\partial q_3}\hat{\mathbf{e}}_3$$

OR

$$\nabla\phi(q_1, q_2, q_3) = \sum_{i=1}^{3} \frac{1}{h_i}\frac{\partial \phi}{\partial q_i}\hat{\mathbf{e}}_i.$$

2 mark

Here, h_i is the scale factor corresponding to the coordinate $\left(h_i = \left|\frac{\partial \mathbf{r}}{\partial q_i}\right|\right)$

$\frac{1}{2}$ **mark**

$\hat{e}_i$ is the unit vector along the q_i coordinate direction. $\left(\hat{\mathbf{e}}_i = \frac{1}{h_i}\frac{\partial \mathbf{r}}{\partial q_i}\right)$

$\frac{1}{2}$ **mark**

9. **Question:-** $\vec{\mathbf{a}} = x^2\hat{i} + yz\hat{j} + z^3\hat{k}$ find $\nabla(\nabla \cdot \vec{a})$
Answer:-

$$\nabla \cdot \vec{\mathbf{a}} = \left(\hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}\right) \cdot \left(x^2\hat{\mathbf{i}} + yz\hat{\mathbf{j}} + z^3\hat{\mathbf{k}}\right) = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(yz) + \frac{\partial}{\partial z}(z^3) = 2x + z + 3z^2$$

$1\frac{1}{2}$ **mark**

$$\nabla(\nabla \cdot \vec{\mathbf{a}}) = \frac{\partial}{\partial x}(2x + z + 3z^2)\hat{\mathbf{i}} + \frac{\partial}{\partial y}(2x + z + 3z^2)\hat{\mathbf{j}} + \frac{\partial}{\partial z}(2x + z + 3z^2)\hat{\mathbf{k}} = 2\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + (1 + 6z)\hat{\mathbf{k}} = 2\hat{\mathbf{i}} + (1 + 6z)\hat{\mathbf{k}}$$

$1\frac{1}{2}$ **mark**

10. **Question:-** State Stoke's theorem?
Answer:- Stoke's theorem relates the surface integral of the curl of a vector field over a surface S to the line integral of the vector field around the boundary curve C of that surface.

OR

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

The circulation of a vector field **A** around a closed curve C is equal to the surface integral of the curl of **A** over any surface **S** bounded by C.

Correct/Sensible Answer - 3 mark

11. Write the integral form of divergence of a vector field over a volume.

$$\int_V (\nabla \cdot \mathbf{A})\, dV = \oint_S \mathbf{A} \cdot d\mathbf{S}$$

$2\frac{1}{2}$ **mark**

The total divergence of a vector field **A** inside a volume V equals the net outward flux through its boundary surface S .i.e. Sum of all local sources inside V = Total outflow through **S**.

$\frac{1}{2}$ **mark**

12. **Question:-** Define the surface integral of a scalar field and a vector field.
Answer:- **Surface Integral of a Scalar Field:**

If $\phi(x, y, z)$ is a scalar field over a surface S, then

$$\int_S \phi\, d\mathbf{S}$$

$1\frac{1}{2}$ **mark**

Surface Integral of a Vector Field:

If $\mathbf{A}(x, y, z)$ is a vector field, its surface integral is

$$\int_S \mathbf{A} \cdot d\mathbf{S}$$

$1\frac{1}{2}$ mark

Part B

Answer any Eight, Each carries 5 marks **(8×5=40)**

13. **Question:-** Explain the concept of basis vectors. Distinguish between orthogonal and non-orthogonal basis vectors.

Answer:- In a vector space, a set of basis vectors is a collection of linearly independent vectors that can represent any vector in the space as a unique linear combination: $\mathbf{A} = a_1\hat{\mathbf{e}}_1 + a_2\hat{\mathbf{e}}_2 + a_3\hat{\mathbf{e}}_3$ where $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$ are the basis vectors and a_1, a_2, a_3 are the components (coordinates) of $\mathbf{A}$.

Or any other Correct/Sensible Answer - 2 mark

Orthogonal Basis Vectors Basis vectors are said to be orthogonal then $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = 0 \quad \text{for } i \neq j$ If each vector also has unit magnitude ($|\hat{\mathbf{e}}_i| = 1$), the basis is orthonormal $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \delta_{ij}$

Or any other Correct/Sensible Answer - $1\frac{1}{2}$ mark

Non-Orthogonal Basis Vectors A set of linearly independent vectors that span a vector space but they ar non-orthogonal that is $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j \neq 0 \quad \text{for some } i \neq j$

Or any other Correct/Sensible Answer - $1\frac{1}{2}$ mark

14. **Question:-** Show that $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}]$

Answer:- The scalar triple product of three vectors is defined as $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ and can be written in determinant form as

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} = \begin{vmatrix} b_x & b_y & b_z \\ c_x & c_y & c_z \\ a_x & a_y & a_z \end{vmatrix}$$

This determinant represents the volume of the parallelepiped formed by the vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$. Hence, $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}] = [\mathbf{c}, \mathbf{a}, \mathbf{b}]$. Therefore the scalar triple product is invariant under cyclic permutation.

OR

The scalar triple product ($[\mathbf{a}, \mathbf{b}, \mathbf{c}]$) can be written as the determinant of a (3×3) matrix whose rows (or columns) are the components of the three vectors ($\mathbf{a}, \mathbf{b}$, and $\mathbf{c}$). Now, if we move the first row to the last and shift the others up that is, perform a cyclic change of rows-the value of a determinant does not change.

Therefore, when we cyclically rearrange the vectors as ($[\mathbf{b}, \mathbf{c}, \mathbf{a}]$), the value of the determinant remains the same. Hence, $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}]$ because cyclic permutations of rows (or vectors) do not affect the determinant's value.

Or any other Correct/Sensible Answer - 5 mark

15. **Question:-** Show that the eigen values of a Hermitian matrix are real.

Answer:- Let $\mathbf{A}$ is a Hermitian Matrix/Operator and $\mathbf{X}^i$ be the set of its Eigen vector.

$$\begin{aligned} \mathbf{AX^i} &= \lambda_i \mathbf{X^i} \\ \left(\mathbf{AX^i}\right)^\dagger &= \left(\lambda_i \mathbf{X^i}\right)^\dagger \\ \mathbf{X^i}^\dagger \mathbf{A}^\dagger &= \lambda_i^* \mathbf{X^i}^\dagger \end{aligned}$$

Multiplying this on the right by $\mathbf{X}^j$ gives

$$\mathbf{X^i}^\dagger \mathbf{A}^\dagger \mathbf{X^j} = \lambda_i^* \mathbf{X^i}^\dagger \mathbf{X^j} \qquad (1)$$

$$\text{Similarly} \quad \mathbf{AX^j} = \lambda_j \mathbf{X^j}$$

Multiplying this on the left by $\left(\mathbf{X}^i\right)^\dagger$ gives

$$\mathbf{X^i}^\dagger \mathbf{AX^j} = \lambda_j \mathbf{X^i}^\dagger \mathbf{X^j} \qquad (2)$$

Eq-(1)-Eq-(2)

$$\begin{aligned} \mathbf{X^i}^\dagger \mathbf{A}^\dagger \mathbf{X^j} - \mathbf{X^i}^\dagger \mathbf{AX^j} &= \lambda_i^* \mathbf{X^i}^\dagger \mathbf{X^j} - \lambda_j \mathbf{X^i}^\dagger \mathbf{X^j} \quad \text{For a Hermitian Matrix/Operator} \quad \mathbf{A}^\dagger = \mathbf{A} \\ 0 &= \left(\lambda_i^* - \lambda_j\right) \mathbf{X^i}^\dagger \mathbf{X^j} \quad \therefore \left(\lambda_i^* - \lambda_j\right) = 0 \quad \textbf{OR} \quad \mathbf{X^i}^\dagger \mathbf{X^j} = 0 \end{aligned}$$

For a nonzero vector with $i = j$. $\mathbf{X^i}^\dagger \mathbf{X^i} \neq 0$

$\therefore \left(\lambda_i^* - \lambda_i\right) = 0$, This is possible only if $\lambda_i^* = \lambda_i$, That is the Eigen value is real

Or any other Correct/Sensible Answer - 5 mark

NB:

one mark can be given for the correct definition of a Hermitian matrix.

one mark can be given for the definition of eigenvalues.

16. **Question:-** Prove that Pauli spin matrices are Hermitian.The Pauli spin matrices are:

$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Answer:- A matrix M is Hermitian if $M^\dagger = M$, where $M^\dagger$ is the *conjugate transpose* of M(i.e., take the transpose and then the complex conjugate of each element).

Or any other Correct/Sensible Answer - 2 mark

(1) For $\sigma_x : \sigma_x^\dagger = (\sigma_x^T)^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x$. Hence, σ_x is Hermitian.

1 mark

(2) For $\sigma_y : \sigma_y^T = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$, $(\sigma_y^T)^* = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_y$. Hence, σ_y is Hermitian.

1 mark

(3) For $\sigma_z : \sigma_z^\dagger = (\sigma_z^T)^* = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z$. Hence, σ_z is Hermitian.

1 mark

17. **Question:-** $\mathbf{A}=\begin{pmatrix}7 & 0 & 6\\ 0 & -5 & 0\\ 6 & 0 & -2\end{pmatrix}$

(a) What is degenerate eigenvalue.Explain,

(b) Show that the matrix A has a degenerate eigenvalue

Answer:- **Degenerate Eigenvalue:** If more than one linearly independent eigenvector correspond to the same eigenvalue, that eigenvalue is called degenerate. $A\mathbf{X}=\lambda\mathbf{X}, \quad A\mathbf{Y}=\lambda\mathbf{Y}, \quad$ with $\mathbf{X}\neq k\mathbf{Y}$. Thus, the same eigenvalue λ is associated with two or more eigenvectors.

Correct/Sensible Answer - $2\frac{1}{2}$ mark

$$\mathbf{A}=\begin{pmatrix}7 & 0 & 6\\ 0 & -5 & 0\\ 6 & 0 & -2\end{pmatrix}$$

Characteristic equation: $\det(\mathbf{A}-\lambda\mathbf{I})=0$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}7-\lambda & 0 & 6\\ 0 & -5-\lambda & 0\\ 6 & 0 & -2-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\begin{vmatrix}7-\lambda & 0 & 6\\ 0 & -5-\lambda & 0\\ 6 & 0 & -2-\lambda\end{vmatrix}=(7-\lambda)[(-5-\lambda)(-2-\lambda)-0]+6[0-6(-5-\lambda)]$$

$$=(-5-\lambda)\left[(7-\lambda)(-2-\lambda)-36\right]=(-5-\lambda)(\lambda^2-5\lambda-50)$$

Solve $(-5-\lambda)(\lambda^2-5\lambda-50)=0$ $\lambda=-5, \quad \lambda^2-5\lambda-50=0 \Longrightarrow \lambda=\dfrac{5\pm\sqrt{225}}{2}=-5,10$.
The Eigenvalues are : $\lambda_1=-5,\lambda_2=-5,\lambda_3=10$

$2\frac{1}{2}$ mark

18. **Question:-** The pressure in a fluid is given by $P(x,y)=x^2y$. Find ∇P and explain its significance.
Answer:-

$P(x,y)=x^2y$. The Gradient of P, $\nabla P=\frac{\partial P}{\partial x}\hat{\mathbf{i}}+\frac{\partial P}{\partial y}\hat{\mathbf{j}}+\frac{\partial P}{\partial z}\hat{\mathbf{k}}=2xy\,\hat{\mathbf{i}}+x^2\hat{\mathbf{j}}$

3 mark

∇P points in the direction of maximum pressure increase. Its magnitude indicates the rate of change of pressure.

2 mark

19. **Question:-** For a rigid body in motion with linear velocity $\mathbf{v}=\omega\times\mathbf{r}$, where $\omega=\omega_x\hat{\mathbf{i}}+\omega_y\hat{\mathbf{j}}+\omega_z\hat{\mathbf{k}}$ and $\mathbf{r}=x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}}$. Show that $\nabla\times\mathbf{v}=\nabla\times\omega\times\mathbf{r}=2\omega$
Answer:-

$$\mathbf{v}=\boldsymbol{\omega}\times\mathbf{r}=\begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ \omega_x & \omega_y & \omega_z\\ x & y & z\end{vmatrix}=(\omega_y z-\omega_z y)\hat{\mathbf{i}}+(\omega_z x-\omega_x z)\hat{\mathbf{j}}+(\omega_x y-\omega_y x)\hat{\mathbf{k}}.$$

$2\frac{1}{2}$ mark

$$\nabla \times (\boldsymbol{\omega} \times \mathbf{r}) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial_x & \partial_y & \partial_z \\ \omega_y z - \omega_z y & \omega_z x - \omega_x z & \omega_x y - \omega_y x \end{vmatrix} = 2(\omega_x \hat{\mathbf{i}} + \omega_y \hat{\mathbf{j}} + \omega_z \hat{\mathbf{k}}) = 2\boldsymbol{\omega}.$$

$$\therefore \quad \nabla \times \mathbf{v} = \nabla \times (\boldsymbol{\omega} \times \mathbf{r}) = 2\boldsymbol{\omega}.$$

$2\frac{1}{2}$ mark

20. **Question:-** A particle moves along a curve defined by $\mathbf{r}(t) = t^2\,\mathbf{i} + 2t\,\mathbf{j} + t^3\,\mathbf{k}$. Find the velocity and acceleration vectors at $t = 1$.

Answer:-

The position vector: $\mathbf{r}(t) = t^2\,\mathbf{i} + 2t\,\mathbf{j} + t^3\,\mathbf{k}$.

$$\text{Velocity:} \quad \mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = 2t\,\mathbf{i} + 2\,\mathbf{j} + 3t^2\,\mathbf{k}.$$

$$\text{Acceleration::} \quad \mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = 2\,\mathbf{i} + 0\,\mathbf{j} + 6t\,\mathbf{k}.$$

3 mark

$$\text{At } t = 1: \quad \mathbf{v}(1) = 2\,\mathbf{i} + 2\,\mathbf{j} + 3\,\mathbf{k}, \quad \mathbf{a}(1) = 2\,\mathbf{i} + 0\,\mathbf{j} + 6\,\mathbf{k}.$$

2 mark

21. **Question:-** $U = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ Find $\mathbf{U^{-1}}$ and verify$\mathbf{U^\dagger U = I}$

Answer:-

$$U = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$U^{-1} = \frac{1}{\det(U)} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{The determinant} \quad \det(U) = (0)(0) - (1)(-1) = 1$$

3 mark

$$U^T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad U \text{is real}, \therefore U^\dagger = U^T \quad U^\dagger = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$U^\dagger U = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$U^{-1} = U^\dagger \quad \text{and} \quad U^\dagger U = I$$

2 mark

22. **Question:-** A force acts along a straight line according to the law $F(x) = x^3 - 3x(N)$. Determine the net work done as a particle moves from $x = -2$ to $x = 2$.

Answer:-

$$W = \int_{-2}^{2} F(x)\,dx = \int_{-2}^{2} (x^3 - 3x)\,dx = \left[\frac{x^4}{4} - \frac{3x^2}{2}\right]_{-2}^{2} = (4 - 6) - (4 - 6) = 0\,\text{J}.$$

Answer with unit - 5 mark